



SpaGAN: A spatially-aware generative adversarial network for building generalization in image maps

Zhiyong Zhou ^a, Cheng Fu ^{b,c,a,*}, Robert Weibel ^a

^a Department of Geography, University of Zurich, Zurich, Switzerland

^b College of Information and Electrical Engineering, China Agricultural University, Beijing, China

^c Key Laboratory of Agricultural Machinery Monitoring and Big Data Applications, Ministry of Agriculture and Rural Affairs, Beijing, China

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ABSTRACT

Building generalization is an essential task in generating multi-scale topographic maps. The progress of deep learning offers a new paradigm to overcome the coordination challenges faced by conventional building generalization algorithms. Some studies have confirmed the feasibility of several original semantic segmentation networks, such as U-Net and its variants and the conditional generative adversarial network (cGAN), for building generalization in image maps. However, they suffer from critical deformation effects, especially for large and geometrically complex buildings. Since learning building generalization essentially means modeling the subtle transformation of building footprints across scales, we argue that the spatial awareness of a neural network, for instance, regarding building size and shape, is crucial to effective learning. Thus, we propose a spatially-aware generative adversarial network, SpaGAN. It takes a representative cGAN, pix2pix, as the backbone, and modifies two modules: In the U-Net-based generator, an atrous spatial pyramid pooling (ASPP) module replaces the conventional convolutional module to extract multi-scale features of buildings of varying sizes and shapes; in the PatchGAN-based discriminator, a signed distance map (SDM) module is used to capture the fine-grained shape difference for discrimination. The proposed network was comprehensively evaluated with a synthetic and a real-world dataset. The results demonstrate that SpaGAN outperforms existing baseline models (U-Net, ResU-Net, pix2pix) for building generalization, particularly in the real-world dataset. The new model can achieve more reasonable aggregation, simplification, and squaring generalization operators.

1. Introduction

Map generalization, which acts as an intellectual activity to structure, select, and represent geographical features (Brassel and Weibel, 1988; Weibel and Bottenfield, 1992; Weibel and Dutton, 1999; Mackaness, 2007), is an essential process for making multi-scale maps in cartography. To enable the automation of such a process, traditional map generalization algorithms mainly formalize cartographic principles and guidelines (Weibel, 1997; Sester, 2005; Buchin et al., 2016; Samsonov and Yakimova, 2017) or constraints (Barrault et al., 2001; Ruas and Duchêne, 2007; Harrie and Weibel, 2007). However, as a topographic map commonly contains diverse and complex geographical features and spatial patterns, formalizing generalization rules and constraints is labor-intensive, if not impossible, for expert cartographers, and it is challenging to coherently coordinate the interplay of these elements in a joint, holistic generalization process.

The recent progress of deep learning (DL) has introduced a new paradigm to address the existing challenges in automated map generalization (Touya et al., 2019; Kang et al., 2024; Harrie et al., 2024).

The most straightforward method is to take image maps (i.e., rasterized maps) as input and model map generalization from large to smaller scales as an image transformation task so that those semantic segmentation networks can be directly adapted to the pipeline. Some early studies have empirically investigated the feasibility of several classical neural networks, including U-Net, generative adversarial networks (GAN), and their variants, in building generalizations (Feng et al., 2019; Kang et al., 2020).

Unfortunately, the generalized buildings from these networks suffer from critical deformation effects in their boundaries (Feng et al., 2019). To mitigate this effect, Fu et al. (2024a) adopted contextual modeling of geographical layers in a tensor (Courtial et al., 2022) and proposed a layered data model that separates the focus building to be generalized from its surrounding buildings into two channels of a tensor. As a result, a neural network can explicitly pay more attention to buildings that should be generalized. Moreover, as deformation effects are considerably inherited from the raster format of image maps,

* Corresponding author at: Department of Geography, University of Zurich, Zurich, Switzerland.

E-mail addresses: cheng.fu@cau.edu.cn, cheng.fu@geo.uzh.ch (C. Fu).

some researchers argue to employ vector maps as input for building generalization (Knura, 2021; Zhou et al., 2022; Yan and Yang, 2024).

Despite improving the smoothness and straightness of generalized buildings of normal size and regular shapes (Fu et al., 2024a), existing approaches perform poorly in generalizing buildings of varying sizes and shapes, particularly for large and geometrically complex buildings, and the deformation bottleneck of DL models remains. On the one hand, varying sizes and shapes of buildings in a map make it difficult for a neural network to learn ‘one size fits all’ weights. On the other hand, few DL models encode the fine-grained spatial information of individual buildings (e.g., shape, size) and groups of buildings (e.g., proximity; Karsznia and Weibel 2018) during their optimization process. This leads to poor awareness of the spatial context in recognizing the geometric characteristics of buildings as objects and the relative positions between buildings (Simmons, 2003; Cheng et al., 2019). However, map generalization, including building generalization, is commonly sensitive to the spatial information of map features, which contextualizes the generalization operations (Mackaness et al., 2014).

Several studies have attempted to address this bottleneck by customizing loss functions regarding spatial information. For instance, Kang et al. (2020) proposed a geometric transformation loss to encode the cartographic knowledge in a CycleGAN model. However, their specified rotation and vertical flip losses show limited effectiveness in generalizing buildings. Courtial et al. (2023) introduced the connectivity of roads to a customized GAN loss for generalizing mountain roads, but their model mainly works on the generalization of linear geographic features instead of polygonal objects like buildings. Theoretically, GANs can be more effective and robust in generating realistic images because they include the discriminator in addition to the generator. However, Feng et al. (2019) found that the original GAN models did not show superior performance over sole generators such as residual U-Net (Zhang et al., 2018), short ResU-Net.

Therefore, it is unclear how spatial information (e.g., building sizes and shapes, and proximity between buildings) can be explicitly encoded to mitigate the deformation bottleneck existing in building generalization of image maps. In this research, we propose **SpaGAN**, a **Spatially-aware Generative Adversarial Network** for building generalization with image maps. It takes a representative conditional generative adversarial network (cGAN), pix2pix (Isola et al., 2017), as the backbone, which is mainly inspired by the fact that cGAN is particularly suitable for supervised learning for paired images, that is, source scale image maps and target scale image maps in our research. The generator of SpaGAN is a U-Net and the discriminator is a PatchGAN.

Notably, in a vector representation, spatial awareness (e.g., regarding size, shape, proximity) among buildings can be directly learned from the representation learning of the vector polygons and has thus been applied to the *classification* of building shape and function in several studies (Yan et al., 2021; Mai et al., 2022; Yu et al., 2024). However, as building generalization involves a series of operators, such as simplification, aggregation, typification, etc. (Stanislawski et al., 2014; Senn et al., 2024), applying the representation learning of vector polygons to building generalization requires the explicit modeling of individual operators as well as the generalization process (e.g., chaining of operators), which tremendously increases the complexity of learning; too far to reach an end-to-end solution. Hence, in this research, we focus on a more straightforward generalization solution with image maps, where spatial awareness is mainly implicitly encoded.

This research makes the following contributions:

- Conceptually, we argue that the spatial awareness of DL models, that is, to accurately recognize the geometrical characteristics of buildings (e.g., their size and shape) and their relative positions (e.g., proximity), is crucial to effective learning of building generalization, where subtle geometric transformations take place across scales, while most prior studies model map generalization in image maps as a mere semantic segmentation task, without geometric transformation.
- Methodologically, SpaGAN incorporates an atrous spatial pyramid pooling (ASPP) module into the generator to enhance its capacity to extract multi-scale features of buildings of varying sizes and shapes for building generalization.
- Unlike many studies solely improving the capacity of the generator in a GAN, this research introduces a signed distance map (SDM) module to the discriminator to capture fine-grained shape differences by encoding its distance to the boundary and distinguishing the inner from the outer part of a building.
- In addition to the synthetic dataset (Feng et al., 2019), this study for the first time introduces a real-world dataset from a national mapping agency to test the capability of deep learning approaches in building generalization. A series of experiments with two datasets comprehensively demonstrate the superior performance of SpaGAN over existing models for building generalization in image maps.

2. Related work

2.1. Constraints-based building generalization

Since buildings are one of the most fundamental features in topographic maps, building generalization has gained particular attention from researchers and mapping agencies (Brassel and Weibel, 1988; Mackaness et al., 2014). Various generalization operators, such as selection, aggregation, and simplification, have been conceived to reduce the number of buildings, ensure their readability, and maintain relationships among buildings (Regnauld and McMaster, 2007; Stanislawski et al., 2014).

In building generalization as a partial process of the overall map generalization process, the most challenging part is to coordinate these operators coherently to make legible maps at multiple scales. To date, constraint-based approaches have shown the most satisfying capability of such coordination in map generalization (Taillandier and Gaffuri, 2012). The core components of these approaches include constraints that define the requirements of generalized maps at a certain scale, as well as optimization algorithms that force the operators to meet the constraints. Expert cartographers have developed mainly six categories of constraints, including position, topology, shape, structural, functional, and legibility (Ruas and Duchêne, 2007) for individual and groups of map objects, while each category of constraints can be further specified by properties such as *area*, *width*, *squareness*, *orientation*, or *alignment* (Hoarau, 2011; Bailey et al., 2014). Common optimization algorithms include agent modeling (Barrault et al., 2001; Ruas and Duchêne, 2007; Regnauld et al., 2014), genetic algorithms (Ware et al., 2003), and least-squares adjustment (Sester, 2005).

However, constraint-based approaches are heavily tailored to the expert-developed procedures that regulate what operators should be applied to which conditions (Harrie and Weibel, 2007). It is challenging for expert cartographers to enumerate and formalize all possible conditions due to the complexity of the geographical reality and its representation on maps (Regnauld et al., 2014). Additionally, managing a large number of generalization algorithms used in agent modeling is also demanding from an engineering perspective.

2.2. Building generalization using deep learning

The striking advance of deep learning (DL) is bringing a new paradigm to map generalization (Touya et al., 2019), promising to reduce the reliance on expert cartographers and knowledge formalization and thus provide end-to-end solutions. To date, building generalization has received the most attention because of its dominance in topographic maps, while some studies are focusing on other geographical features, such as roads (Courtial et al., 2023), coastlines (Du et al., 2022), and contour lines (Yu and Chen, 2022). The mainstream DL

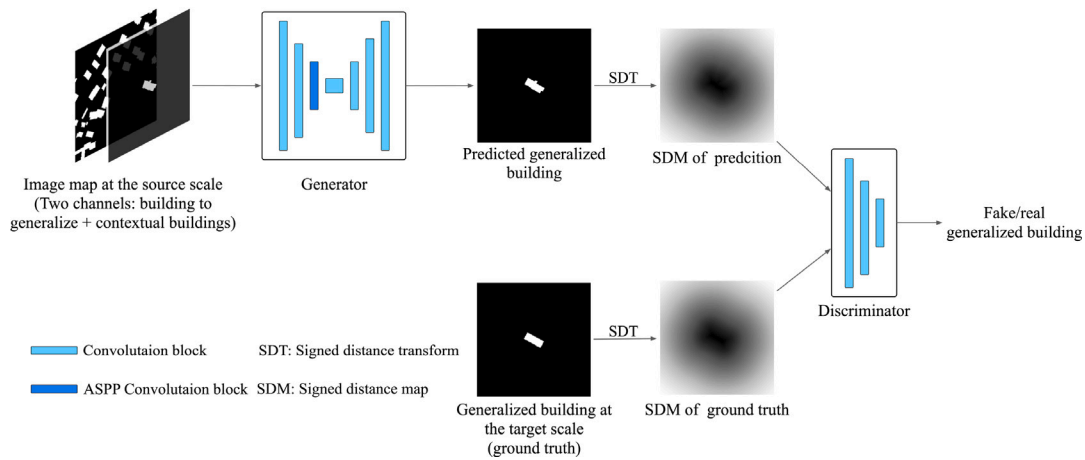


Fig. 1. The proposed SpaGAN architecture. ©swisstopo.

solution for building generalization is to take image maps as input and output and to consider the generation of target-scale image maps as analogous to an image transformation task (Feng et al., 2019; Kang et al., 2020). Unfortunately, there exists a severe deformation effect on the generated target-scale image maps. Fu et al. (2024a) mitigated such an issue by proposing a multi-channel data model to separate the generalized building from its surrounding buildings for learning. However, building generalization knowledge about the coordination of a set of operators (e.g., simplification after aggregating a group of small buildings) has not yet been fully learned.

Considering the inherent noise from the image format of maps, it is an intuitive way to adopt vector maps as input to mitigate the deformation effect in DL-based building generalization. Currently, graph convolutional neural networks are frequently used to encode building polygons and learn building features in vector maps (Yan et al., 2019). Regarding the downstream map generalization task, Zhou et al. (2023) formulated the building simplification operator as a joint task of removing and moving vertices of building polygon footprints and introduced a multi-task learning model to enable a graph convolutional neural network to simplify buildings. Yan and Yang (2024) introduced an autoencoder architecture for building simplification. However, these vector-based DL approaches can only perform simplification of individual buildings but no other operators involving multiple buildings, such as aggregation.

2.3. Spatially-aware semantic segmentation

As introduced in Section 2.1, the constraints in map generalization mainly work on the spatial contexts of individuals and groups of map objects. This inspires us to investigate spatially-aware learning, which is an important approach to improving semantic segmentation performance. A set of neural network operations have been proposed to capture multi-scale features of objects (i.e., varying sizes and shapes), including spatial transformer networks (Jaderberg et al., 2015), atrous spatial pyramid pooling (ASPP; Chen et al. 2017a), or coordinate convolution (CoordConv, Liu et al. 2018).

Moreover, many studies differentiate the pixel functionality in shaping the boundary of segmentation based on the distance to the boundary pixels and the inner or outer part relative to the boundary (Ma et al., 2020). Considering the heavy computational resources required for the pixel-by-pixel distance calculation, the distance map is more frequently used to encode the distance of each pixel relative to its nearest boundary of an object and thus can quantify the loss compared to the object boundaries (Hayder et al., 2017; Caliva et al., 2019; Xue et al., 2020). Furthermore, combining the boundary-based loss and region loss into a cross-entropy loss has also been shown an advantage over solely using the boundary loss (Kervadec et al., 2019).

In the domain of geographic information science, some studies have noticed the importance of boundaries in spatially-aware semantic segmentation. For example, Bokhovkin and Burnaev (2019) and Yang et al. (2023) customized the boundary-based losses for remote sensing image segmentation. To capture the continuity of map features of varying shapes and sizes, Wu et al. (2022) integrated an ASPP module to the U-Net for digitalizing historical maps, which may provide inspiration regarding the extraction of multi-scale building features in the context of deep learning for building generalization. However, it should be noted that all of those spatially-aware semantic segmentation approaches are solely dedicated to segmenting a single object, without geometric transformation. In contrast, building generalization from large to small scales typically involves geometric transformations over multiple buildings (e.g., in an aggregation operation), which is a more complicated process than semantic segmentation. Thus, the capacity of existing spatially-aware semantic segmentation approaches for building generalization remains unknown. Moreover, these approaches mainly focus on the improvement of generators, but the spatial awareness of GAN models receives little attention.

3. SpaGAN model

3.1. Problem definition

Similar to the previous studies on DL-based building generalization in image maps, this research also equates the building generalization from large-scale to small-scale maps with a semantic segmentation problem. To better understand and illustrate the domain complexity of building generalization, this research defines the problem of building generalization in image maps as follows:

Given a binary image map at a source scale and the corresponding binary image map at a target scale, respectively, where each pixel is assigned with *is_building* labels (typically with a value 1) or *not_building* labels (typically assigned as 0), the main goal of learning building generalization is to learn the geometric transformation of buildings from the source scale to the target scale, which includes various generalization operators (e.g., aggregation, displacement, simplification, etc.).

3.2. Overall architecture

Having identified the lack of explicit spatial information encoding for building generalization using deep learning, this research proposes SpaGAN, a spatially-aware generative adversarial network for building generalization in image maps. It aims to effectively learn generalization knowledge, particularly involving large and geometrically complex

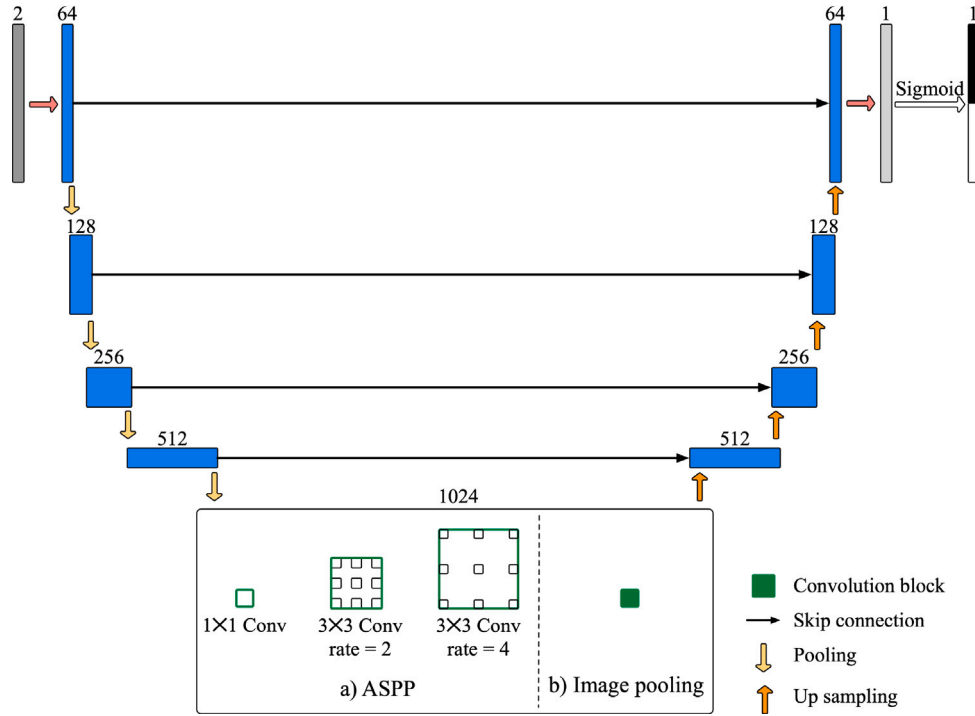


Fig. 2. The architecture of the U-Net generator with ASPP.

buildings, and produce a more realistic target-scale map from a given source-scale map.

Fig. 1 illustrates the architecture of SpaGAN, which follows the design of the pix2pix model for paired images (Isola et al., 2017). As input, an image map at the source scale, which is represented by a building-centered two-channel data model, is fed into the network. In the two-channel data model, a building to generalize is centered in the first channel, and its contextual buildings are represented in the second channel. Choosing such a data model allows the explicit control of the learning focus of a neural network and thus improves its learning efficiency (Fu et al., 2024a). The generator of SpaGAN is a U-Net architecture, but we add an atrous spatial pyramid pooling module (ASPP) following the encoder as an additional bridge to enable more effective feature extraction for buildings of varying sizes and shapes (see Section 3.3). Subsequently, instead of directly being fed into a discriminator to discriminate between fake or real, the predicted generalized binary building image and the ground truth building image are first transformed into signed distance maps (SDMs) by a signed distance transform (SDT) function, which can enrich the spatial information for the discrimination (see Section 3.4). Finally, the discriminator can predict if the fed SDMs are from fake or real generalized buildings.

3.3. U-Net generator with atrous spatial pyramid pooling

As shown in Fig. 2, the proposed SpaGAN adopts the U-Net architecture as the generator. Since the buildings in image maps have very distinct sizes and shapes, conventional U-Net models have a limited capacity to capture the multiple scales of features of objects because their receptive fields solely focus on the local scale (Lv et al., 2022). For this reason, we introduce an atrous spatial pyramid pooling (ASPP) module to the U-Net architecture (see the bottom layer of Fig. 2). Following the design in Chen et al. (2017b), the ASPP module consists of one 1×1 convolution and two 3×3 atrous convolutions with rates 2 and 4, and a global image pooling layer.

3.4. Signed distance map-based discriminator

The discriminator of SpaGAN is a PatchGAN-based architecture that discriminates whether small patches of images are fake or real (Isola et al., 2017). However, as our image maps are binary-encoded, the original PatchGAN is very likely to fail to differentiate the predicted and ground-truth generalized building images by capturing their fine-grained differences' interior. For example, misclassified boundary pixels in a building's interior may not cause a significant difference in the global classification accuracy but it matters significantly to form the shape of the building (Bokhovkin and Burnaev, 2019). Additionally, for building generalization operations involving multiple buildings, the interior and exterior of a building should also contribute unevenly (Stanislawski et al., 2014). For instance, in the aggregation operation, the exterior pixels between two close buildings will be converted to the inner part of a building merged from the two source buildings, which means that these exterior pixels should receive other pixels to learn this operation.

Inspired by the above-mentioned intuition and relevant studies in computer vision (Kervadec et al., 2019; Xue et al., 2020), we propose transforming the binary image map to a signed distance map (SDM) and then feeding the SDM to the PatchGAN-based discriminator. As shown in Fig. 1, each pixel can be more easily distinguished based on the SDM than in a binary image map because the SDM encodes the spatial information of each pixel relative to the boundary pixels, including the relative distance and directions (inner or outer). An SDM results from the signed distance transform (SDT). In this research, the SDT of an image map M is defined as:

$$M_{SDT} = \begin{cases} -\inf_{y \in \partial_M} \|x - y\|_2, & x \in M_{in} \\ 0, & x \in \partial_M \\ \inf_{y \in \partial_M} \|x - y\|_2, & x \in M_{out} \end{cases} \quad (1)$$

where $\|x - y\|_2$ is the Euclidean distance from any pixel x to the closest boundary pixel y , ∂_M denotes the set of pixels representing boundaries of buildings in an image map, and M_{in} and M_{out} denote the inner and outer parts of buildings, respectively.

4. Experiments and evaluation

We conducted a series of experiments with two multi-scale map datasets from two different sources of different characteristics, with the primary objective of investigating how the proposed SpaGAN performs compared to existing models for building generalization in image maps.

4.1. Experimental setup

4.1.1. Datasets

In our experiments we used two datasets, a synthetic dataset, called OSM-Stuttgart, and a real-world dataset, called swisstopo-Zurich:

- (1) *OSM-Stuttgart dataset*: The dataset was constructed by [Feng et al. \(2019\)](#) extracting buildings from the crowd-sourced mapping platform OpenStreetMap (OSM) and generalizing these to the target scales 1:5000, 1:10,000 and 1:25,000 using the conventional map generalization software *CHANGE* ([Powitz, 1993](#)), which mainly applied simplification of individual building footprints and aggregation among buildings. Since the dataset lacks the interactive interventions and corrections of expert cartographers, it can be regarded as a synthetic dataset that is relatively easy to learn for DL models. The resolution of the raw rasterized vector maps is $0.5 \text{ m} \times 0.5 \text{ m}$. In our experiments, we used the source scale of 1:5000 and the target scale of 1:10,000. [Table 1](#) shows the training, validation, and test set sizes. Notably, the test set is from another area of Stuttgart that has no overlap with the areas of the training and validation sets. Thus, its number of buildings does not follow a certain ratio of the training and validation sets.
- (2) *swisstopo-Zurich dataset*: The buildings contained in this dataset were produced by professional cartographers at swisstopo at two scales, 1:10,000 and 1:25,000, with the interactive assistance of automated map generalization algorithms. The buildings of the two separate scales were then matched via a unique user ID (UUID) and geometric matching to produce a multi-scale dataset, with 1:10,000 denoting the source scale and 1:25,000 the target scale. The resolution of the rasterized image maps in the dataset is set to $0.875 \text{ m} \times 0.875 \text{ m}$.¹ We chose buildings in the canton of Zurich, Switzerland, for training and validation and buildings in the cantons of Basel and Lucerne for testing. As this is a real-world dataset, there are a large number of small-area buildings and non-generalized buildings, which makes the dataset highly imbalanced for learning building generalization regarding the generalization complexity. Additionally, a lower Intersection over Union (IoU) between source and target buildings (*src_tar_iou*) commonly indicates higher uncertainty about the involved operations (e.g., matching errors might have occurred after the displacement of buildings). These may cause a more severe imbalance in the building generalization complexity for training. Therefore, we excluded buildings with areas smaller than 100 pixels (small buildings), and *src_tar_iou* larger than 0.98 (non-generalized buildings) and smaller than 0.5 (uncertain buildings). As a result, 80,299 (small: 25,669; non-generalized: 43,255; uncertain: 11,375) out of 208,399 buildings were excluded from the training and validation sets, and 2021 (small: 1301; uncertain: 1620) out of 13,861 buildings were excluded from the test set. Notably, we did not exclude non-generalized buildings in the test set to evaluate the generality of the trained models. The resulting numbers of the dataset are reported in [Table 1](#).

¹ The resolution derives from the multiplication between swisstopo-regulated minimum side length of a rectangle building, 0.35 mm, and the scale of 1:25,000.

Table 1

Size of datasets used in the experiments.

	Training	Validation	Test
OSM-Stuttgart (1:5,000 → 1:10,000)	85,819	9,535	1,545
swisstopo-Zurich (1:10,000 → 25,000)	115,290	12,810	11,812

Table 2

Quantiles of OSM-Stuttgart and Swisstopo-Zurich datasets.

Dataset part	Quantiles	<i>src_tar_iou</i> (OSM-Stuttgart)	<i>src_tar_iou</i> (swisstopo-Zurich)
Training set	0.2	0.951	0.753
	0.4	0.997	0.833
	0.6	0.999	0.887
	0.8	1.000	0.931
Test set	0.2	0.942	0.639
	0.4	0.997	0.729
	0.6	0.999	0.837
	0.8	1.000	0.952

Additionally, we quantitatively compared the characteristics of these two datasets based on *src_tar_iou*, which can be further used to demonstrate the capacity of the proposed model. The lower *src_tar_iou* is, the more generalized a map is. We plotted the empirical cumulative distribution (eCD) of *src_tar_iou*s of the training set and test set in the OSM-Stuttgart and Swisstopo-Zurich datasets, respectively. As shown in [Fig. 3](#), 80% of data points of the OSM-Stuttgart training set have a *src_tar_iou* of 0.95 and higher, while those in the swisstopo-Zurich training set are 0.75 and above. The difference between the two datasets also exists in the test set part. This indicates that the real-world swisstopo-Zurich dataset has a higher complexity of generalization for buildings than the synthetic OSM-Stuttgart dataset. Thus, swisstopo-Zurich is a more complex and challenging dataset compared to OSM-Stuttgart. Such characteristics can also be verified through numerical values of quantiles as reported in [Table 2](#).

4.1.2. Implementation details

The proposed SpaGAN was implemented in Pytorch 3.9 and trained with a single NVIDIA V100 Tensor Core GPU. The optimizers for the generator and discriminator are both Adam optimizers, with a batch size of 8. The loss functions in SpaGAN are consistent with the original pix2pix model ([Isola et al., 2017](#)). Specifically, the generator adopts a combination of L2 loss and L1 loss, while the discriminator employs L2 loss. The learning rate for training the SpaGAN mode was set to 2×10^{-4} in the OSM-Stuttgart dataset and 4×10^{-4} in the swisstopo-Zurich dataset, respectively. To avoid over-fitting and save computational cost, we employed early stopping with the criterion that the performance (F1 score) of the trained models does not improve over three epochs (same as [Feng et al. 2019](#)).

4.1.3. Evaluation metrics

To achieve a comprehensive evaluation of the proposed SpaGAN's performance, we propose to use a richer set of metrics for quantitative evaluation than previous, related studies ([Feng et al., 2019](#); [Kang et al., 2020](#)):

- (1) Intersection over Union (IoU): As the ratio of the intersection of pixels to the union of pixels between prediction and ground truth, this measure is commonly used to evaluate the pixel-level performance in image generation tasks.
- (2) Boundary Intersection over Union (BIOU): Considering that the shapes of generalized buildings are a key factor influencing map legibility ([Balley et al., 2014](#)) and the shape of a building is dominated by its boundary, BIOU is adopted along with the following BDis measure as metrics that focus on the evaluation of contour shapes ([Li et al., 2024](#)). BIOU is easily implemented,

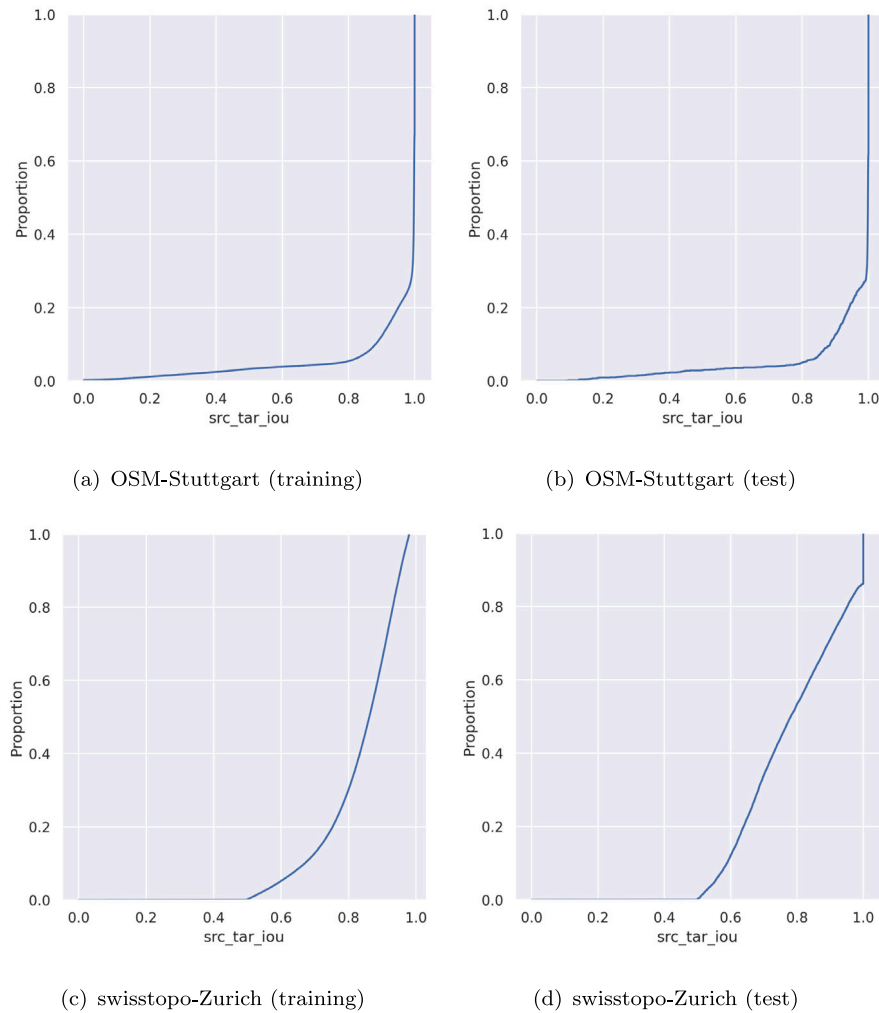


Fig. 3. Empirical cumulative distributions (eCD) of the *src_tar_iou* of OSM-Stuttgart and swisstopo-Zurich datasets.

as its only difference from IoU is using boundary pixels as inputs. Notably, it is the first time that boundary-based metrics are introduced to evaluate DL-based raster map generalization products.

- (3) Boundary distance (BDis): Since BIoU results from the binary evaluation of each boundary pixel, it is unable to capture the finer-grained shape difference between predicted and ground-truth generalized buildings. Therefore, we introduce another distance-based metric, BDis. It is calculated as: $BDis = DIF_{boundary} \cdot M_{gt}$, where $DIF_{boundary}$ denotes the pixel set of the boundary difference between the prediction and ground truth, and M_{gt} is the non-signed distance map of a ground-truth building. Notably, as distant pixels contribute less to the building shape, we use the reciprocal Euclidean distance of each pixel to the nearest boundary pixel to create the distance map (Caliva et al., 2019).
- (4) Rectangularity index (RI): This is the ratio between the area of a polygon and the area of its minimum area bounding rectangle (Basaraner and Cetinkaya, 2017), which measures the regularity of generalized building shapes.
- (5) Fractality index (FI): Measures the edge roughness of boundaries, which is calculated as $1 - \frac{\log A}{2 \cdot \log P}$, where A denotes the area of a building, and P is the perimeter of the building.

- (6) Smoothness index (SI): Similar to FI, this metric also evaluates the roughness or smoothness of boundaries. Proposed by Basaraner and Cetinkaya (2017) under the name “roughness index”, it should be less sensitive to elongation and more responsive to roughness than the FI. It is calculated as $\frac{r^2}{A + P^2} \cdot 42.62$, where r denotes the average length of the radial lines between the centroid of a building and the vertices of its boundary. Since higher values of this metric indicate better performance in terms of smoothness, we use the term *smoothness index* (SI) rather than the original term, *roughness index*.

We distinguish two types of evaluations. The evaluations with IoU, BIoU, and BDis are termed *external evaluations* since these metrics are derived from the comparison with the ground-truth of generalized maps at the target scale, while the evaluations using RI, FI, and SI are termed *internal evaluations*, as they are calculated based on their own geometric characteristics.

4.2. Quantitative evaluations

As prior studies (Feng et al., 2019; Fu et al., 2024a) have investigated, ResU-Net and GAN models such as pix2pix have promising capacities for building generalization in image maps. Therefore, the ResU-Net model and the pix2pix model are chosen as the baselines

Table 3

Results of external quantitative evaluation. ↑ means that the higher a value is, the better the performance is for a particular metric, while ↓ means the opposite. Bold numbers indicate the best result for a given metric, and underlined numbers refer to the second best model. The top 50% of test cases regarding the high complexity of generalization (i.e., src_tar_iou below the median value) is denoted by H50%, while the lower half is denoted by L50%, referring to those cases having a lower complexity of generalization.

Dataset	Model	IoU (↑)			BioU (↑)			BDis (↓)		
		Overall	H50%	L50%	Overall	H50%	L50%	Overall	H50%	L50%
OSM-Stuttgart	U-Net	0.9802	0.9685	0.9920	0.9347	0.8808	0.9886	0.0210	0.0388	0.0031
	ResU-Net	0.9788	0.9643	0.9934	0.9333	0.8761	0.9907	<u>0.0210</u>	0.0396	<u>0.0024</u>
	pix2pix	0.9624	0.9318	0.9931	0.8875	0.7852	0.9897	0.0345	0.0681	0.0010
	SpaGAN	0.9810	0.9688	<u>0.9933</u>	0.9361	0.8821	<u>0.9900</u>	0.0204	0.0380	0.0027
swisstopo-Zurich	U-Net	<u>0.8039</u>	0.6729	<u>0.9349</u>	0.4773	<u>0.2034</u>	<u>0.7511</u>	0.1759	0.2439	0.1079
	ResU-Net	0.8030	<u>0.6718</u>	0.9340	0.4759	0.2016	0.7502	<u>0.1678</u>	<u>0.2364</u>	<u>0.0992</u>
	pix2pix	0.7772	0.6616	0.8928	0.4077	0.1923	0.6230	0.1849	0.2440	0.1258
	SpaGAN	0.8044	0.6705	0.9382	0.4851	0.2050	0.7652	0.1670	0.2362	0.0978

to evaluate SpaGAN. Moreover, U-Net, the generator of SpaGAN and pix2pix, is also taken as a baseline model to identify how an adversarial network architecture works for map generalization. However, another commonly used U-Net variant, Attention U-Net, is not included in the baseline models because prior studies (Winkler, 2023; Fu et al., 2024a) have shown that the attention mechanism has a limited capacity to improve the performance in building generalization when a two-channel data model (i.e., focal building in one channel, and contextual buildings in the other channel) is used as in this paper to purposely force a neural network to focus on the central areas where the focal building should be generalized.

Sections 4.2.1 and 4.2.2 present the results of the external and internal evaluation, respectively. The mean values of the introduced evaluation metrics were taken as the overall performance of each model in the entire test set. In particular, since the data distribution of map generalization is inherently imbalanced and skewed (see Fig. 3), mean values are likely to underestimate or overestimate the performance of the models. Thus, the test sets were divided into two halves by the median value of src_tar_iou , denoted as H50% and L50%, respectively, in the following tables (see the caption of Table 3 for an explanation).

4.2.1. External evaluations

Table 3 shows that in the synthetic OSM-Stuttgart dataset the proposed SpaGAN achieved the best performance regarding the overall evaluations. Its overall IoU is 0.08% higher than the second-best model, U-Net. For BioU, SpaGAN achieved 93.61% in the complete dataset, improving by 0.14%, 0.28%, and 4.86% from the U-Net, ResU-Net, and pix2pix, respectively. Additionally, the BDis of SpaGAN is 0.0204, being reduced by 2.86% compared to the second-best models, either U-Net or ResU-Net. Regarding the evaluations by generalization complexity (H50% and L50%), while SpaGAN surpassed all of the other three models in the dataset part of higher complexity (H50%), its performance in the dataset part of lower complexity (L50%) is slightly lower than the best results. Specifically, the IoU (0.9933) and BioU (0.9900) of SpaGAN are 0.01% and 0.07% lower than the best ResU-Net model, respectively, and the BDis of SpaGAN is 0.0027, 0.0017 larger than the best result from pix2pix.

By comparison, it is more evident that SpaGAN outperformed the other three baseline models in the real-world swisstopo-Zurich dataset, which contains a larger number of buildings and more geometrically complicated buildings than the OSM-Stuttgart dataset. Furthermore, in this dataset, a greater variation of generalization operators has been applied. For IoU, SpaGAN achieved an overall IoU of 80.44%, increasing by 0.05% compared with U-Net. However, in the H50% and L50% parts of the dataset, the relative performance of these two models shows a contradictory result: for H50%, U-Net is 0.24% higher; for L50%, SpaGAN is 0.33% higher. In terms of the boundary-based metrics, SpaGAN resulted in an overall BioU of 48.51%, an improvement of 0.78% (H50%: 0.16%; L50%: 1.41%) over U-Net. Moreover, the BDis

of SpaGAN drops by 0.48% from the value of 0.1678 in the second-best model, ResU-Net. In the H50% part, SpaGAN saw a BDis reduction of 0.08% compared to ResU-Net, while the decrease in the L50% part is more substantial, about 1.41%.

To avoid the overestimation or underestimation of model performance brought by the skewed distribution of test results, we performed the Wilcoxon signed-rank test, which is a non-parametric test method, for the paired predictions between SpaGAN and any one of the baseline models, respectively. As shown in Table 4, in the OSM-Stuttgart dataset, the performance of SpaGAN has no significant difference from U-Net and ResU-Net in terms of IoU and BioU (but it does differ significantly from pix2pix), while the median BDis of U-Net is even significantly lower than that of SpaGAN. In contrast, SpaGAN significantly outperformed U-Net, ResU-Net and pix2pix in the swisstopo-Zurich dataset, with significantly higher IoU and BioU, and significantly lower BDis.

4.2.2. Internal evaluations

The results of the quantitative internal evaluation in terms of RI, FI, and SI are presented in Table 5. In the OSM-Stuttgart dataset, the RI (0.8754) of SpaGAN is generally better than the other three baseline models, 0.0003, 0.0005, and 0.0104 higher than U-Net, ResU-Net, and pix2pix, respectively. In the H50% part, SpaGAN resulted in an RI of 0.8357, 0.0017, 0.0020, and 0.0206 higher than U-Net, ResU-Net, and pix2pix, respectively. However, in the L50% part, U-Net had the highest RI, 0.9165, followed by ResU-Net with a value of 0.9157. Regarding FI, the best overall value (0.3227) and the H50% value (0.3207) both resulted from the pix2pix model, whereas in the L50% part, the ResU-Net performed best, with a value of 0.3215. Moreover, SpaGAN achieved the highest overall SI (0.7188) and H50% (0.6892), 0.0012, and 0.0031 better than the second-best result, respectively. In the L50% part, pix2pix got the best SI, 0.7559.

In the swisstopo-Zurich dataset, the relative performance among the four models is similar to that in the OSM-Stuttgart dataset. The RI of SpaGAN in the complete dataset (0.8366), H50% (0.8230), and L50% (0.8501), respectively, are all superior to the other models. The results for the FI show that U-Net obtained values of 0.3487 overall, 0.3529 in H50%, and 0.03446 in L50%, respectively, which are 0.0005, 0.0007, and 0.0002 lower (and thus better) than the second-best model, U-Net. Likewise, ResU-Net ranked as the second-best regarding the SI metric, 0.0066, 0.0077, and 0.0069 lower than the best model, pix2pix.

The non-parametric test results for the internal evaluation metrics are reported in Table 6. The results indicate that the proposed SpaGAN model improved RI significantly compared to the U-Net and pix2pix models in both datasets. However, it did not outperform the ResU-Net in the OSM-Stuttgart dataset. In terms of FI, only in the swisstopo-Zurich dataset, SpaGAN's performance was significantly better than pix2pix. Moreover, regarding SI, SpaGAN achieved only significantly higher (and thus better) values than pix2pix in both datasets. While there is no significant difference between SpaGAN and U-Net as well

Table 4

Results for the Wilcoxon signed-rank test of external evaluation metrics. ‘+’ denotes that the median value of the former model is higher than the latter model, while ‘-’ denotes lower. * indicates a significance level of 0.05, and *** indicates a significance level of 0.001. Bold numbers indicate that SpaGAN significantly performs better regarding the metric.

Metrics	OSM-Stuttgart			swisstopo-Zurich		
	U-Net - SpaGAN	ResU-Net - SpaGAN	pix2pix - SpaGAN	U-Net - SpaGAN	ResU-Net - SpaGAN	pix2pix - SpaGAN
IoU (↑)	+	+	_***	_***	_***	_***
BloU (↑)	+	+	_***	_***	_***	_***
BDIs (↓)	_*	-	+***	+***	+***	+***

Table 5

Results of internal quantitative evaluation. For the meaning column titles, see Table 3.

Dataset	Model	RI(↑)			FI (↓)			SI (↑)		
		Overall	H50%	L50%	Overall	H50%	L50%	Overall	H50%	L50%
OSM-Stuttgart	U-Net	0.8751	0.8337	0.9165	0.3247	0.3274	0.3219	0.7176	0.6861	0.7491
	ResU-Net	0.8749	0.8340	0.9157	0.3223	0.3243	0.3215	0.7176	0.6873	0.7479
	pix2pix	0.8650	0.8151	0.9148	0.3227	0.3207	0.3246	0.7109	0.6659	0.7559
	SpaGAN	0.8754	0.8357	0.9151	0.3260	0.3294	0.3225	0.7188	0.6892	0.7483
swisstopo-Zurich	U-Net	0.8319	0.8179	0.8460	0.3487	0.3529	0.3446	0.6863	0.6778	0.6947
	ResU-Net	0.8342	0.8201	0.8483	0.3492	0.3536	0.3448	0.6909	0.6822	0.6995
	pix2pix	0.8063	0.7915	0.8211	0.3634	0.3685	0.3583	0.6975	0.6899	0.7051
	SpaGAN	0.8366	0.8230	0.8501	0.3517	0.3559	0.3475	0.6900	0.6811	0.6990

Table 6

Results for the Wilcoxon signed-rank test of internal evaluation metrics. In addition to the symbols explained in Table 4, ** indicates the significance level of 0.01.

Metrics	OSM-Stuttgart			swisstopo-Zurich		
	U-Net - SpaGAN	ResU-Net - SpaGAN	pix2pix - SpaGAN	U-Net - SpaGAN	ResU-Net - SpaGAN	pix2pix - SpaGAN
RI (↑)	_*	-	_***	_***	_***	_***
FI (↓)	-	_*	-	_***	_***	+***
SI (↑)	-	-	_***	+***	+***	_***

as ResU-Net in the OSM-Stuttgart dataset, the SI of SpaGAN was even significantly worse than U-Net and ResU-Net in the swisstopo-Zurich dataset.

4.3. Visual evaluations

To achieve more intuitive comparisons of the models, we conducted visual evaluations of selected image maps. As the quantitative evaluations in Section 4.2 clearly showed that the pix2pix model is inferior to the ResU-Net and the proposed SpaGAN models while U-Net performed better regarding more metrics than ResU-Net did, we mainly present the visual comparison between SpaGAN and U-Net in addition to the source (i.e., input) and target (i.e., ground truth) image maps.

4.3.1. OSM-Stuttgart dataset

Fig. 4 illustrates the top five cases regarding the generalization complexity, i.e., with the least src_tar_iou values. More sample cases can be found in Figs. B.1 and B.2 of Appendix B. The cases involve a typical generalization procedure, first aggregating buildings and then simplifying the boundary of the merged building by considering its shape rectangularity, and edge orthogonality (Stanislawski et al., 2014). The detailed evaluations are as follows:

- Case 1 shows that our proposed SpaGAN model generated a rectangular building, almost the same as the target building. In contrast, the building generalized by U-Net has a pitched wall at Location *a* compared to the target building and the SpaGAN-generated building.

- Likewise, at Locations *b* (Case 2) and *c* (Case 3), the walls resulting from the U-Net are commonly curved, and less straight than those from SpaGAN.
- In Case 4, to more visibly compare the difference between U-Net and SpaGAN at Location *d*, we zoom in on a particular detail (see the inset ellipses in red). It is clear that both walls of the two buildings have some deformed parts, but the wall generated by SpaGAN is less rugged compared with that generated by U-Net.
- Case 5 demonstrates a more complicated generalization scenario where five surrounding buildings are aggregated with the center building and the resulting merged shape then simplified. Comparing the buildings generalized by U-Net and SpaGAN with the target building at Locations *e* and *f*, our proposed SpaGAN performed a more complete operation with respect to joining the footprints of the individual source buildings regarding their orthogonality and boundary smoothness.

4.3.2. swisstopo-Zurich dataset

Since the top-generalized cases based on src_tar_iou in the OSM-Stuttgart dataset are commonly small-size buildings that have been aggregated with surrounding buildings, we chose five representative cases from the swisstopo-Zurich dataset for the visual comparison instead of the five top-generalized cases as in the OSM-Stuttgart dataset. As shown in Fig. 5, the source buildings have much more complex shapes. More sample cases can be found in Figs. B.3 and B.4 of Appendix B. The detailed evaluations are as follows:

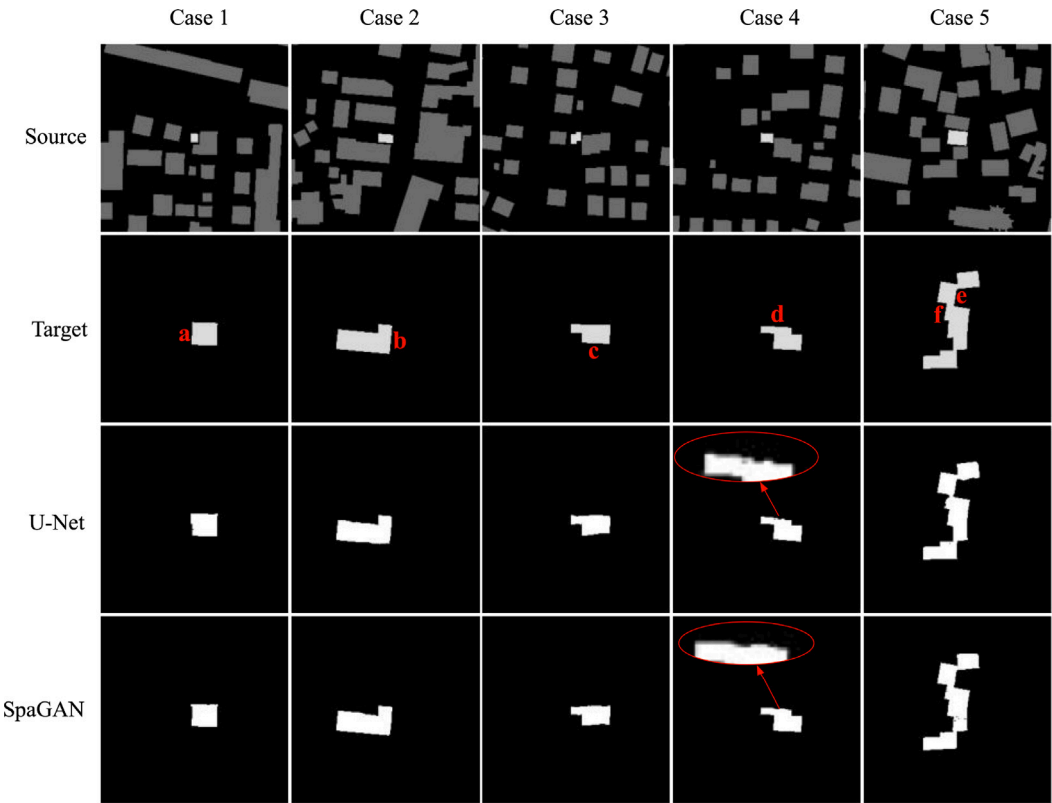


Fig. 4. Visual comparison between the U-Net and SpaGAN models in the OSM-Stuttgart dataset. An image map in the *Source* row shows a building to generalize in the center, highlighted in white, while the darker buildings represent the surroundings of the building to generalize. The corresponding image map in the *Target* row shows solely the generalized building. The buildings in the rows *U-Net* and *SpaGAN* are the corresponding predictions by the two models.

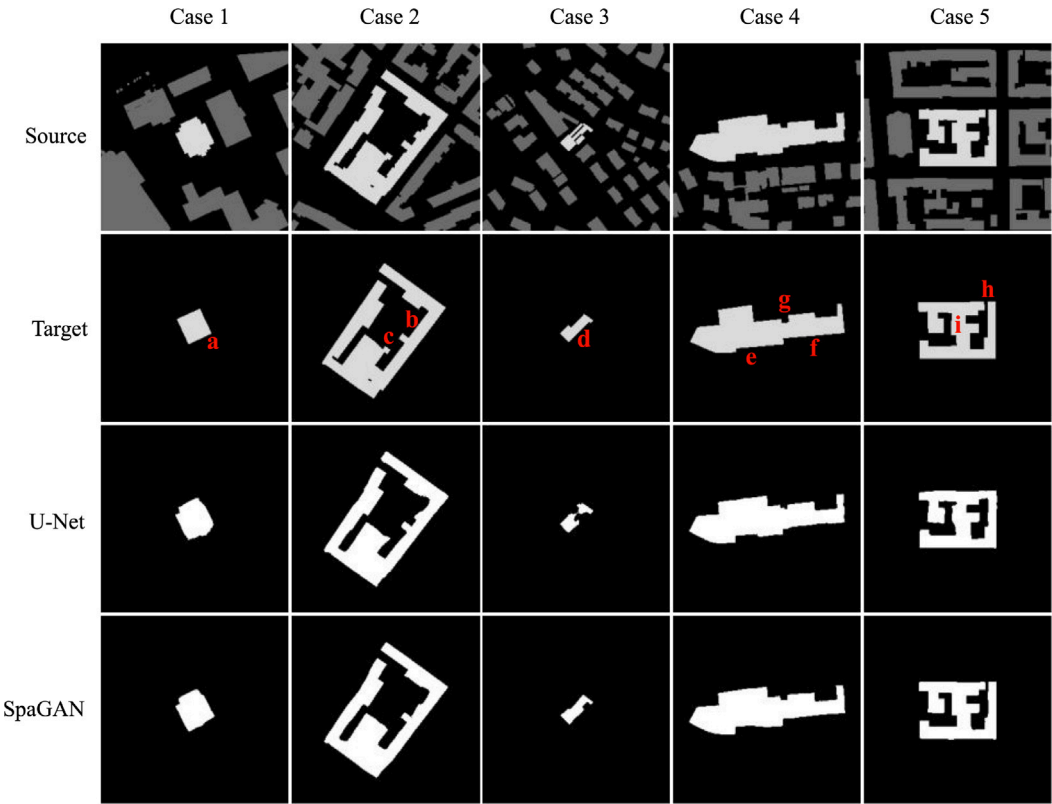


Fig. 5. Visual comparison between the U-Net and SpaGAN models in the swisstopo-Zurich dataset.

- In Case 1, the building is generalized to a rectangle as a typical operation. However, U-Net performed poorly in the bottom right corner of the building (see Location *a*), while SpaGAN achieved a better rectangularity at this location.
- Case 2 showcases the generalization of a ring-shaped building, where the walls are simplified to be more smooth and straight. It shows that an extrusion that should be removed in the target building (see Location *b*) has not been performed by U-Net. In comparison, SpaGAN had a more explicit operation on this part, though a tiny extrusion still remains visible. Additionally, at Location *c*, while U-Net retained the extruded small area, the shape is not as regular as the ground-truth target building. Instead, SpaGAN eroded this extrusion with an orthogonal corner, which makes the generalized building seem more regular.
- In Case 3, there is an aggregation of a building complex from the source to the target image maps. Unfortunately, U-Net almost entirely failed to generalize this case because rather than aggregating the building complex it even separated it into two buildings with irregular shapes. Our proposed SpaGAN generated a more reasonable shape of the generalized building with an intrusion at Location *d* where the wall is straight in the target building.
- Case 4 illustrates the simplification of another large building, where the intruded walls at Locations *e* and *f* are filled in to create smooth, straight walls. Comparing the results of U-Net and SpaGAN, it is evident that SpaGAN performed better in these two intruded locations. Furthermore, there is a tiny intrusion (see Location *g*) that should be kept in the target building. However, U-Net attempted to fill it in while SpaGAN kept it well.
- Case 5 shows another generalization example for a complicated building with irregular shapes, which mainly involves the simplification operator. At Location *h* where a visible gap exists between two parts of the building, U-Net performed aggregation for these two parts, thus obviously violating the topological relationship of this building. In contrast, SpaGAN not only strictly maintained the topology but also resulted in straight walls and orthogonal corners (e.g., at Location *i*).

4.4. Module analysis of SpaGAN

As introduced in Section 3, based on a typical conditional GAN model, pix2pix, the proposed SpaGAN incorporates an atrous spatial pyramid pooling (ASPP) module in the U-Net-based generator and alters the binary image maps input to the PatchGAN with the signed distance map (SDM). To analyze the influence of these two newly introduced modules, we performed ablation studies by removing either one of the two modules in SpaGAN. As a model excluding both ASPP and SDM is equivalent to the pix2pix model, which has been reported in previous sections, we do not include it again in this section. The median values of the numerical results are reported in Table 7.

- W/o ASPP: In the synthetic OSM-Stuttgart dataset, after removing the ASPP module, the performance of the model increases by 12.5% and 0.03% compared to SpaGAN in terms of BDis and FI, respectively. On the contrary, for the BioU, RI, and SI metrics, the performance of the model decreases by 0.01%, 0.01%, and 0.15%, respectively. In contrast, the evaluations with the real-world swisstopo-Zurich dataset show a more consistent performance across the metrics, where all of the metrics deteriorate. Specifically, the IoU, BioU, RI, and SI of w/o ASPP decline by 0.09%, 0.4%, 0.22%, and 0.02%, respectively, while the BDis and FI have increments of 2.00% and 0.05%, respectively.
- W/o SDM: With the removal of the SDM module in the discriminator of SpaGAN, the training loss became NaN in both the OSM-Stuttgart dataset and the swisstopo-Zurich dataset after only

one iteration. Thus, we marked this by “/” in Table 7. Such results show that training a SpaGAN model without the SDM module is prone to collapse, which will be further discussed in Section 5.3.

5. Discussion

5.1. Advantages of SpaGAN for building generalization

Synthesizing the results of the quantitative evaluation (Section 4.2) and the visual evaluation (Section 4.3), we believe it is fair to say that our proposed SpaGAN model improved the performance of building generalization of the existing baseline models, U-Net, ResU-Net and pix2pix. Starting with the results of the OSM-Stuttgart dataset, the non-parametric test results (Table 4) show that in this synthetic dataset, there is no significant difference in IoU and BioU between ResU-Net and SpaGAN, meaning that SpaGAN has a similar overall performance as U-Net and ResU-Net in a dataset of low generalization complexity (see Figs. 3(a) and 3(b) where most of generalization cases stay above 0.95 of *src_tar_iou*). However, the visually superior generalization of buildings (Fig. 4) in the top range of cases regarding generalization complexity and the higher IoU, BioU, and BDis in the H50% class indicate that SpaGAN can indeed improve building generalization compared to the prior U-Net baseline model, even in this dataset that contains fewer complex generalization cases. Furthermore, we generated the scatterplots of all test buildings regarding their resulting evaluation metrics (IoU, BioU, BDis) for both U-Net and SpaGAN. As illustrated in Fig. A.1 of Appendix A, there are more dark dots, which denote a high generalization complexity, situated above the green reference line in subfigure (c) (BioU) and below in subfigure (e) (BDis). Such a distribution further confirms that SpaGAN has a better capacity to generalize these more complex buildings that require generalization.

In the real-world swisstopo-Zurich dataset, the superior performance of SpaGAN over the two baseline models is even more evident. The significantly better performance on all of three external evaluation metrics and one out of three internal evaluation metric (Table 4), together with the metrics distribution of individual test results (Fig. A.1), indicate that SpaGAN can produce building generalization results whose shapes are closer to the generalized buildings in the ground truth. In turn, these results reveal that SpaGAN gains better spatial awareness (e.g., building sizes and shapes) for building generalization compared to U-Net, ResU-Net, and pix2pix. Furthermore, the significant improvement in the rectangularity index (RI) by SpaGAN over both the swisstopo-Zurich and the OSM-Stuttgart datasets (Table 6) also confirms such enhanced spatial awareness. Rectangularity is one of the crucial patterns to be observed in building generalization (Stanislawski et al., 2014; Yan et al., 2017). However, SpaGAN resulted in worse FI and SI than U-Net and ResU-Net. This result can be mainly attributed to SpaGAN making more intensive geometric transformations and focusing more on the boundaries of buildings in the prediction than on interior areas, causing smaller areas and longer perimeters of buildings. This has an effect on both FI, which is reciprocal to the area and proportional to the perimeters, and SI, which is dominantly reciprocal to the power of perimeters, as shown in Section 4.1.3.

Additionally, pix2pix, which was chosen as a representative conditional generative adversarial network, is shown to be inferior to other models. It is consistent with the observation of Feng et al. (2019) that the original GAN discriminators, like PatchGAN of the pix2pix model used in this research, do not necessarily improve the building generalization performance. Nevertheless, SpaGAN and our experimental results support this insight and underscore the feasibility of altering the discriminator in GAN models in building generalization. However, it should be noted that U-Net achieved higher IoU and BioU than ResU-Net in our experiment on the Stuttgart dataset, which contradicts the empirical finding of Feng et al. (2019). This result might be attributed to the difference in the underlying data model (single-channel in Feng

Table 7

Performance comparison between SpaGAN, SpaGAN without ASPP module (w/o ASPP), and SpaGAN without SDM module (w/o SDM), respectively. Median values are shown.

Dataset	Model	IoU (↑)	BloU (↑)	BDis (↓)	RI (↑)	FI (↓)	SI (↑)
OSM-Stuttgart	SpaGAN	0.9983	0.9909	0.0008	0.9209	0.3155	0.7422
	w/o ASPP	0.9983	0.9908	0.0007	0.9199	0.3154	0.7407
	w/o SDM	/	/	/	/	/	/
swisstopo-Zurich	SpaGAN	0.8019	0.3842	0.0850	0.8868	0.3470	0.7140
	w/o ASPP	0.8010	0.3802	0.0867	0.8846	0.3475	0.7138
	w/o SDM	/	/	/	/	/	/

et al. (2019) versus two-channel in this research). More specifically, when a two-channel data model separates the focal building from its contextual buildings, a residual block of ResU-Net may over-estimate the importance of contextual building pixels that are not accounted in the optimization of the model.

5.2. Learned building generalization knowledge

In this research, the OSM-Stuttgart dataset represents the transition from a very large scale (1:5000) to a large scale (1:10,000), while the swisstopo-Zurich dataset involves the transition between two large scales, from 1:10,000 to 1:25,000. The primary generalization operators that take place for buildings in such scale transitions mainly include simplification, enlargement, aggregation, squaring, and displacement (Stanislawski et al., 2014). As shown in Section 4.3 and in Appendix B, it is clear that SpaGAN learned the simplification, aggregation, and squaring operators as well as the coordination among these. For example, the top-complexity cases shown in the Stuttgart dataset (Fig. 4) demonstrate a good capacity to aggregate adjacent buildings. More importantly, our proposed SpaGAN performed well in simplifying the aggregated buildings with high geometric complexity and also achieved more rectangular shapes and straighter edges. This can also be verified by the cases in the swisstopo-Zurich dataset (Fig. 5), where there are more buildings of higher shape complexity with larger and more irregular footprints.

Interestingly, we noticed that some SpaGAN-generalized buildings obtain shapes that represent alternatives to, or seem even more reasonable than, the ground-truth buildings at the target scale. As illustrated in Fig. 6, in the original dataset, source buildings in Cases 1–4 are maintained as individual buildings at the target scale. However, SpaGAN aggregated these individual buildings with their immediate neighbors. Furthermore, in Case 5, the source building is aggregated with a small neighboring building to the right to form a single building at the target scale, but it was kept as an individual building and merely simplified by SpaGAN. Comparing the distances separating the involved source buildings, it is obvious that the distances in Cases 1–4 are smaller than that in Case 5, meaning that based on separation distance, the source buildings in Cases 1–4 should rather be aggregated than in Case 5. It is thus a good sign that our proposed SpaGAN model learned a consistent proximity principle for aggregation in building generalization (Regnauld and McMaster, 2007), which can assist cartographers in identifying and correcting cases inconsistent with generalization principles (in this case, minimum separation distance).

5.3. Influence of ASPP and SDM modules

As the experiment of Section 4.4 showed, the SpaGAN without ASPP led to a decrease in BloU, BDis, and RI metrics relative to SpaGAN in both datasets, indicating that the ASPP module can benefit the model in capturing the fine-grained boundary features of buildings. This confirms our hypothesis that the ASPP module, which is good at extracting

multi-scale features in images (Chen et al., 2017b), fits our building generalization scenario of image maps where buildings commonly vary in size and shape. On the other hand, the performance of the w/o ASPP model (Table 7) had an evident improvement compared with pix2pix (Table 3), exhibiting the positive impact of our proposed SDM module in discriminating generalized buildings that are predicted by the original U-Net model in the pix2pix model.

Moreover, the discrimination power of the SDM module is further underlined through the training collapse of the w/o SDM model on both datasets. It is very likely caused by the generator of the w/o SDM model, which is a U-Net incorporating the ASPP module, generating too realistic buildings so that the discriminator in the model fails to optimize based on its low discrimination capability (Mao et al., 2019; Gui et al., 2021). This can be further verified by the fact that the training of the pix2pix model, which does not incorporate the ASPP module and the SDM module simultaneously, did not collapse. Consequently, we conclude that improving the spatial awareness of a generative adversarial network for building generalization in image maps lies in two aspects: enhancing the capacity of the generator to generate more real fake image maps and strengthening the capacity of the discriminator to distinguish the fine-grained shape differences between the fake and real image maps.

5.4. Limitations

While this research demonstrates that our proposed SpaGAN is capable of performing more reasonable building generalizations in image maps and outperforming the existing baseline models, several limitations can be further improved and addressed in future research.

First, apart from the simplification, aggregation, and squaring operators for building generalization, it is rarely observed that SpaGAN learned other generalization operators, such as the enlargement and displacement of buildings. On the one hand, this may be attributed to the few samples containing these operators, leading to predominance by other primary operators such as simplification and aggregation. Thus, constructing a more balanced dataset is desirable for DL-based building generalization (Harrie et al., 2024), which can also, in turn, lay a more reliable foundation to compare and validate various deep learning models for building generalization. On the other hand, the raster data model, with its 'pixelated' perspective, may also not be appropriate for modeling operators that focus on adjusting the layout and distance relations of buildings globally, such as displacement. Perhaps vector maps and graph convolutional neural networks might offer a more suitable data model and DL architecture (Yan et al., 2019; Yu et al., 2024).

Second, due to the high shape variance of buildings on a map, it is extremely challenging to train a one-size-fits-all model for building generalization. For example, in Case 15 in Fig. B.2, SpaGAN fails to aggregate buildings and produces a worse generalization result than U-Net. Such a challenge can also be observed from the scatter plots in Fig. A.1. Although SpaGAN generally outperforms U-Net regarding

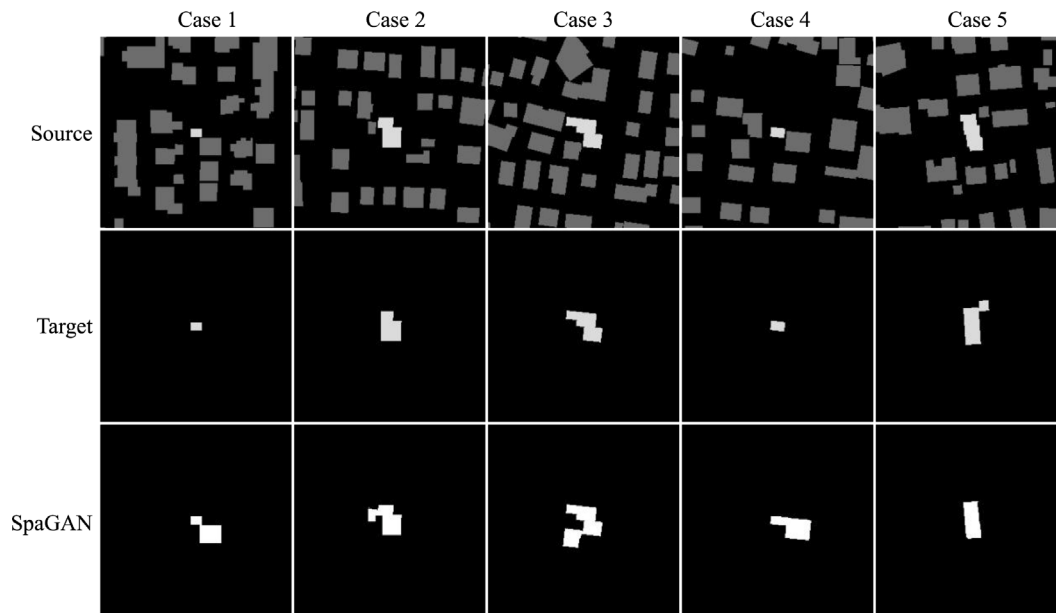


Fig. 6. SpaGAN performed a more reasonable aggregation. .

the evaluation metrics, U-Net can perform well in some cases where SpaGAN performs poorly. Therefore, there is no reason to deny other well-trained models for building generalization. It would be interesting to introduce ensemble methods (Hu et al., 2023) to allow multiple trained models to jointly generalize buildings.

Meanwhile, as illustrated in Fig. A.1, U-Net can perform well in some cases where SpaGAN had relatively lower performance. Therefore, U-Net should also not be ignored to compensate and complement SpaGAN for generalizing many buildings and thus improve the stability of DL-based building generalization. Therefore, it would be interesting to introduce a voting mechanism (Hu et al., 2023) to allow multiple trained models to jointly generalize buildings.

Additionally, the rapid progress in computer vision has given birth to a lot of deep learning models for image transformation. While we mainly focused on improving the spatial awareness of the GAN architecture, comparing the results with its incorporated generator, the U-Net model, it is worth adapting other advanced models (e.g., transformers) to the task of building generalization and further exploring the capacity of deep learning models in building generalization. Nevertheless, it is still recommended to give priority to the boundaries of a building in learning building generalization, as demonstrated in this research as well as in another recent study (Fu et al., 2024b).

6. Conclusions

In this research, we argue that building generalization at large to medium scales, which involves subtle geometric transformations regarding building sizes and shapes across scales, requires DL models to hold a higher capacity of spatial awareness. Therefore, we proposed a spatially-aware generative adversarial network, SpaGAN. Taking a representative conditional generative adversarial network, pix2pix, as a backbone, SpaGAN introduces an atrous spatial pyramid pooling (ASPP) module into the generator and a signed distance map (SDM) module into the discriminator. The former module aims to extract more effective multi-scale features for buildings of various sizes and shapes, while the latter aims to strengthen the discrimination capacity of the GAN. We conducted a series of experiments on a synthetic dataset (OSM-Stuttgart) and a real-world dataset (swisstopo-Zurich).

The results demonstrated that SpaGAN outperformed baseline models, including U-Net, ResU-Net, and pix2pix, in building generalization. Moreover, we identified that SpaGAN effectively learned knowledge about simplification, aggregation, and squaring operators through its production of straighter and more rectangular buildings at target scales.

CRediT authorship contribution statement

Zhiyong Zhou: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cheng Fu:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Robert Weibel:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Individual test results distribution and comparison

See Fig. A.1.

Appendix B. Sample cases of test results

See Figs. B.1–B.4.

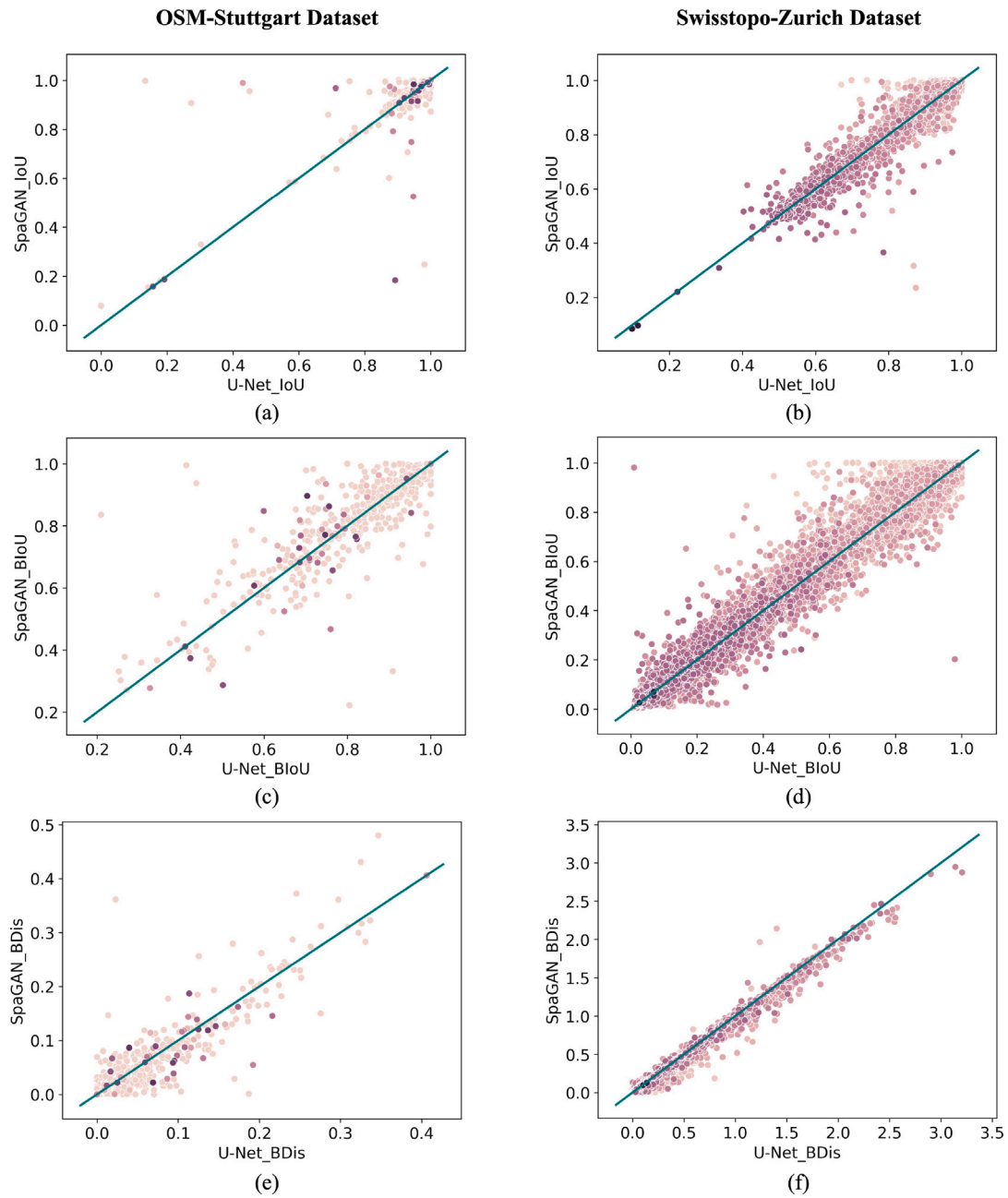


Fig. A.1. Individual test results comparison between the U-Net and SpaGAN models in two datasets, regarding the IoU, BloU, and BDis metrics. The color of dots in figures indicates the complexity of generalization from source image maps to target image maps based on the src_tar_iou metric. The darker dots in the figures denote a higher complexity of generalization. The green reference lines are where the SpaGAN holds the same performance as the U-Net. For IoU and BloU, a dot in the upper part of the reference lines denotes that SpaGAN has a better performance than U-Net, while the lower part means SpaGAN is better in BDis.

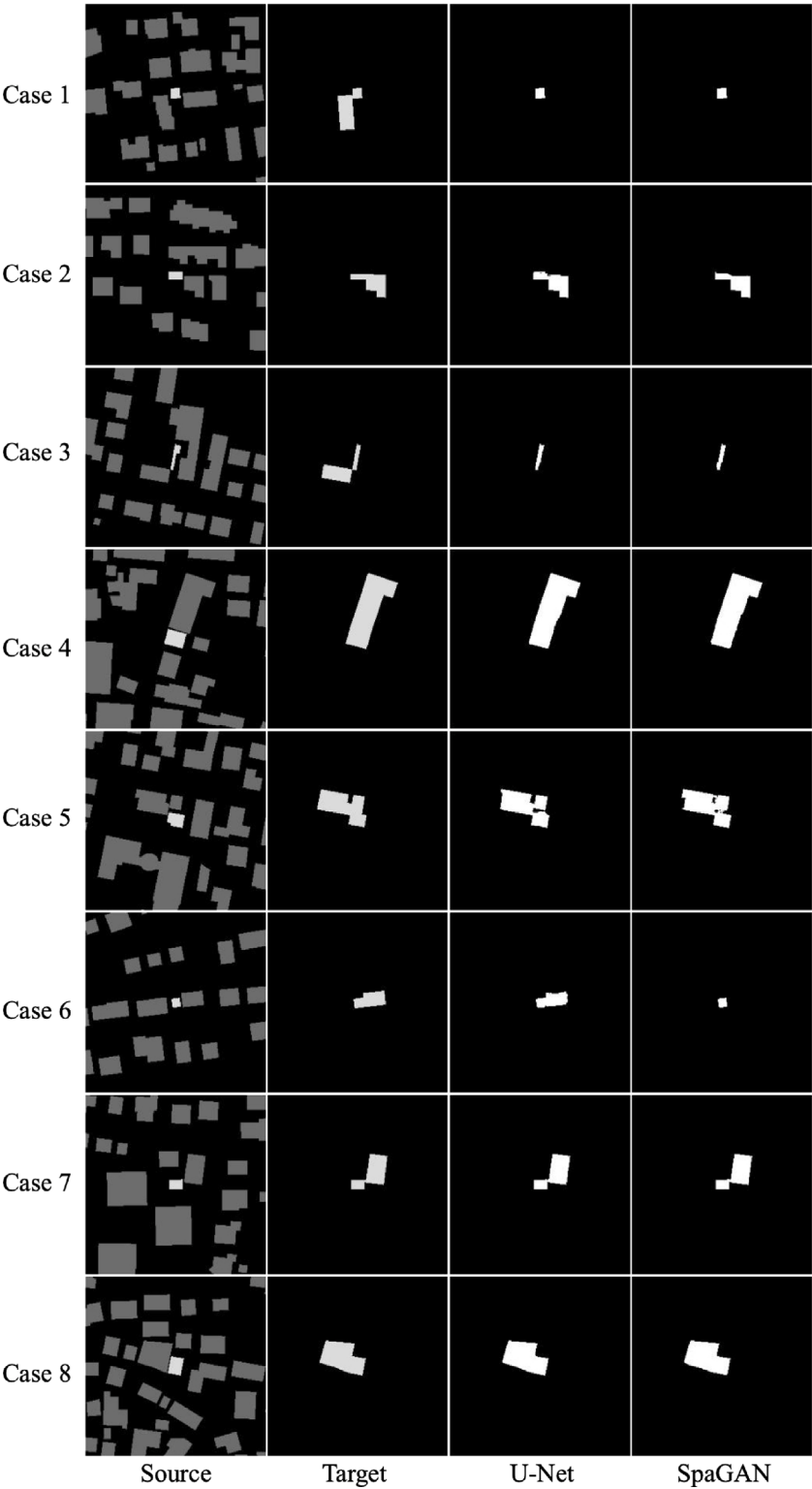


Fig. B.1. Selected test results for the OSM-Stuttgart dataset. [@swisstopo](#).

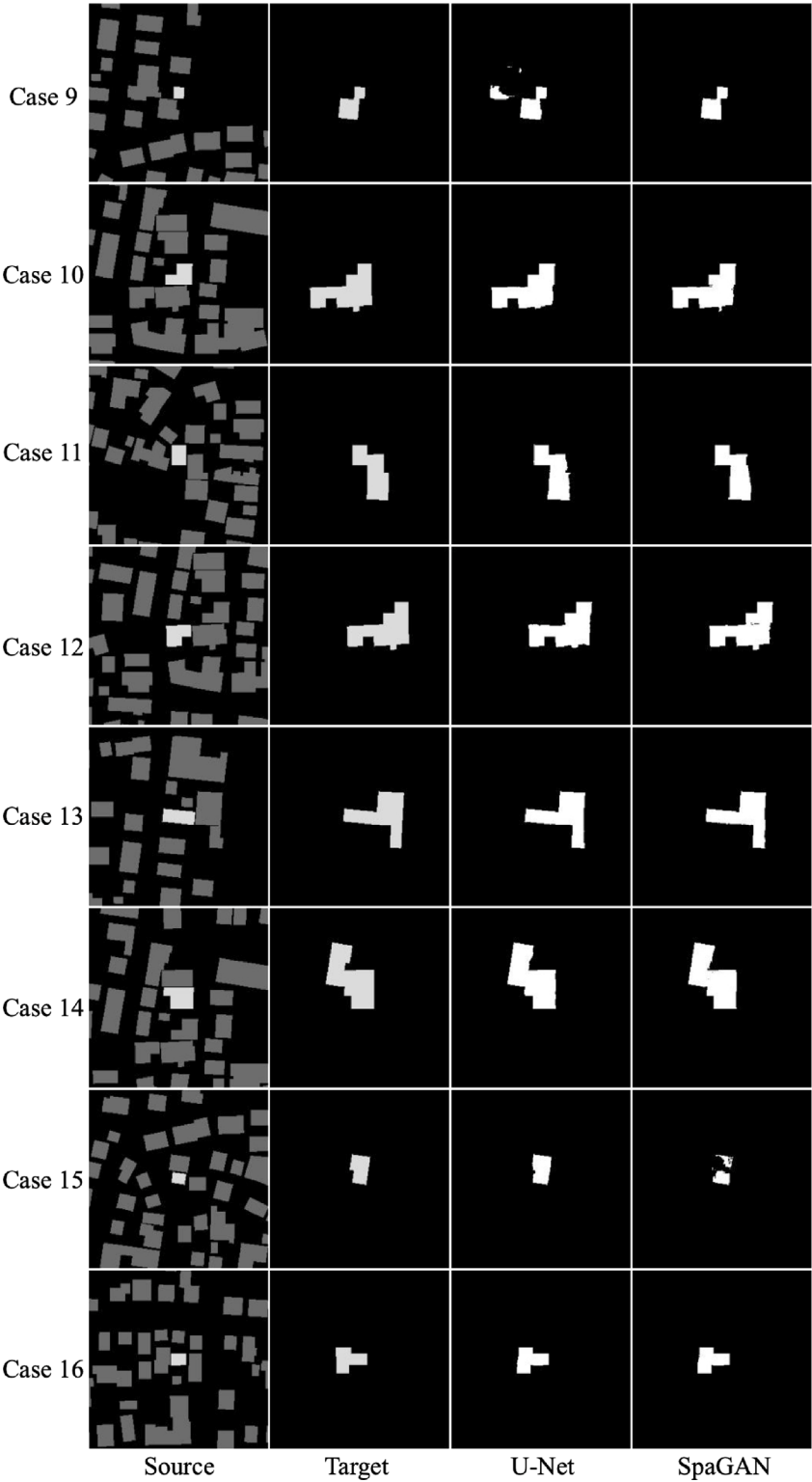


Fig. B.2. Selected test results for the OSM-Stuttgart dataset. [@swisstopo](#).

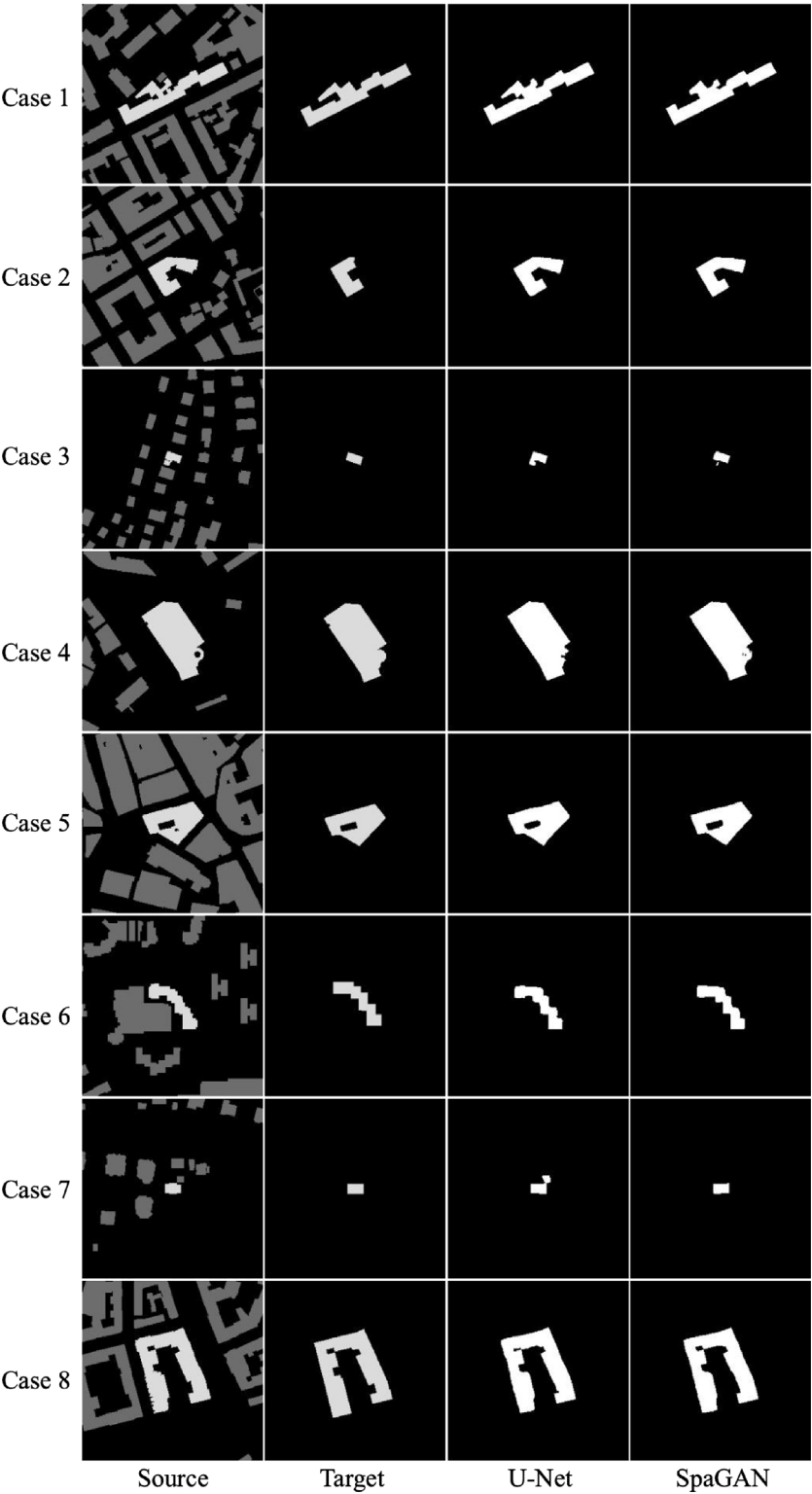


Fig. B.3. Selected test results for the swisstopo-Zurich dataset. [Cswisstopo](#).

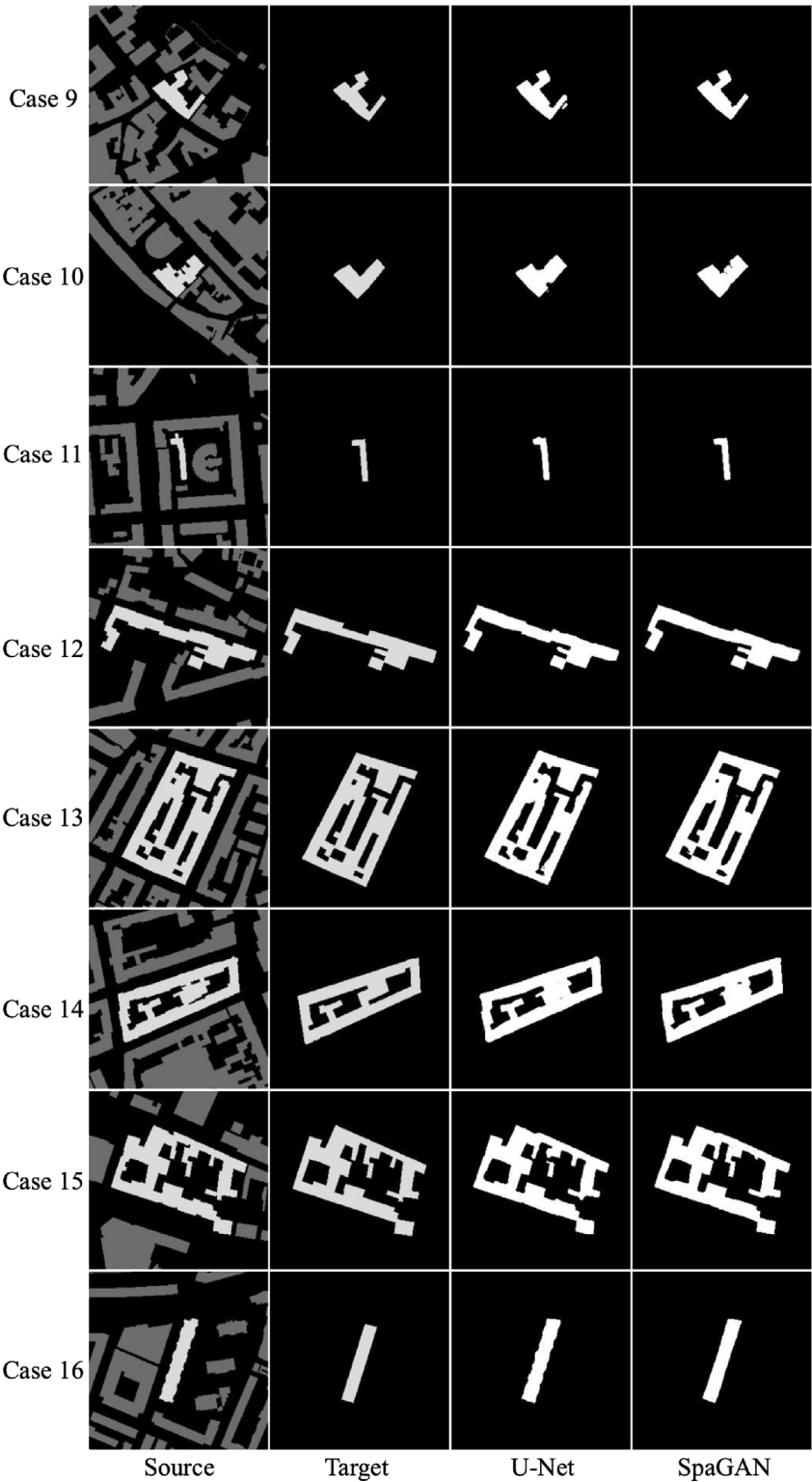


Fig. B.4. Selected test results for the swisstopo-Zurich dataset. [Cswisstopo](#).

Data availability

The authors do not have permission to share data.

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